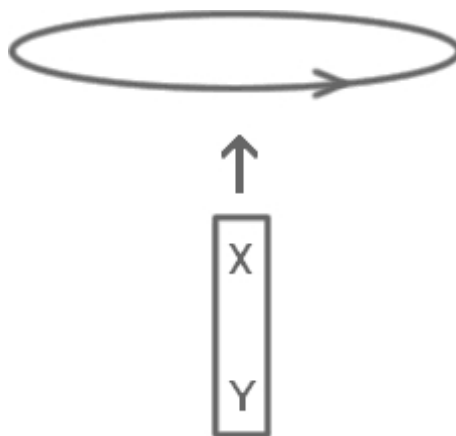


- 1) A deuteron is moving with a speed of $2.0 \times 10^6 \text{ m/s}$ at right angles to a magnetic field. The field is uniform, with magnitude $B = 0.40 \text{ T}$. The mass and charge of a deuteron are $3.3 \times 10^{-27} \text{ kg}$ and $1.6 \times 10^{-19} \text{ C}$, respectively. The radius of the deuteron orbit is approximately (a deuteron is the nucleus of deuterium, an isotope of hydrogen) 1) _____
- A) 6.3 m B) 1.8 m C) 0.21 m D) 10 cm

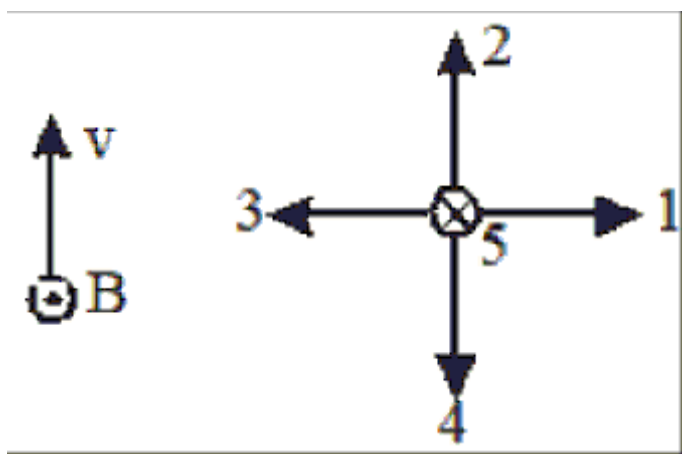
- 2) A flat circular loop of radius 0.10 m is rotating in a uniform magnetic field of 0.20 T. Find the magnetic flux through the loop when the plane of the loop and the magnetic field vector are perpendicular. 2) _____
- A) $0 \text{ T} \cdot \text{m}^2$ B) $3.1 \times 10^{-3} \text{ T} \cdot \text{m}^2$
 C) $5.5 \times 10^{-3} \text{ T} \cdot \text{m}^2$ D) $6.3 \times 10^{-3} \text{ T} \cdot \text{m}^2$

- 3) The figure shows a bar magnet moving vertically upward toward a horizontal coil. The poles of the bar magnets are labeled X and Y. As the bar magnet approaches the coil it induces an electric current in the direction indicated on the figure (counter-clockwise as viewed from above). What are the correct polarities of the magnet? 3) _____



- A) Both X and Y are north poles.
 B) X is a north pole, Y is a south pole.
 C) Both X and Y are south poles.
 D) X is a south pole, Y is a north pole.
 E) The polarities of the magnet cannot be determined from the information given.
- 4) If two elements of a circuit are in parallel, they must have the same 4) _____
- A) current B) voltage across them
 C) resistance D) charge
- 5) A straight wire that is 0.60 m long is carrying a current of 2.0 A. It is placed in a uniform magnetic field of strength 0.30 T. If the wire experiences a force of 0.18 N, what angle does the wire make with respect to the magnetic field? 5) _____
- A) 30° B) 25° C) 90° D) 60° E) 35°

- 6) The left diagram shows a positively charged particle is moving with velocity v in a magnetic field B . Using the right diagram, the direction of the magnetic force on the particle is _____

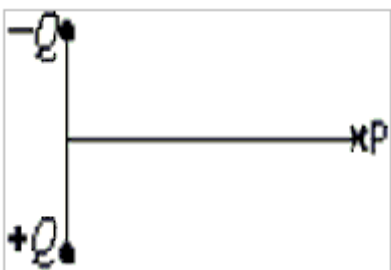


- A) 2 B) 3 C) 1 D) 4

- 7) The coil in a 60-Hz ac generator has 125 turns, each having an area of $3.0 \times 10^{-2} \text{ m}^2$ and is rotated in a uniform 0.12-T magnetic field. What is the peak output voltage of this generator? _____

- A) 220 V B) 200 V C) 120 V D) 110 V E) 170 V

- 8) The diagram shows a particle with positive charge Q and a particle with negative charge $-Q$. The electric field at point P on the perpendicular bisector of the line joining them is: _____
(hint: remember dipoles ?)

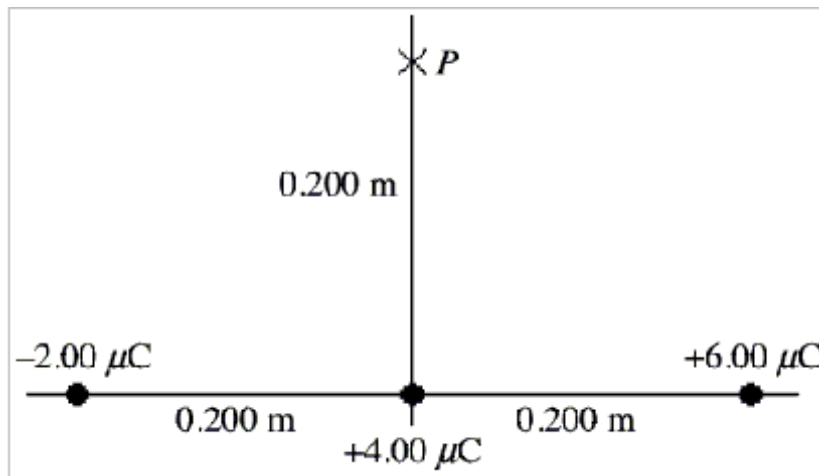


- A) down B) right C) up D) left

- 9) If the charge on a parallel-plate capacitor is doubled: _____
- | | |
|---------------------------------|----------------------------------|
| A) the electric field is halved | B) the capacitance is halved |
| C) the capacitance is doubled | D) the electric field is doubled |

- 10) Three point charges of $-2.00 \mu\text{C}$, $+4.00 \mu\text{C}$, and $+6.00 \mu\text{C}$ are placed along the x-axis as shown in the figure. What is the electrical potential at point P (relative to infinity) due to these charges?

10) _____



- A) 154KV B) -154KV C) +307 kV D) 0

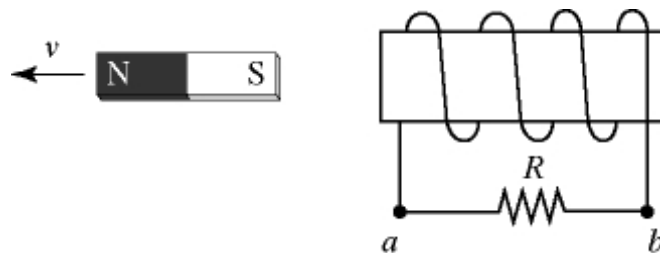
- 11) About 1350 W/m^2 of electromagnetic energy reaches the upper atmosphere of the earth from the sun, which is $1.5 \times 10^{11} \text{ m}$ away. Use this information to estimate the average power output of the sun.

11) _____

- A) $2 \times 10^{26} \text{ W}$ B) $1 \times 10^{26} \text{ W}$ C) $3 \times 10^{26} \text{ W}$ D) $4 \times 10^{26} \text{ W}$

- 12) In the figure, a bar magnet moves away from the solenoid. The induced current through the resistor R is

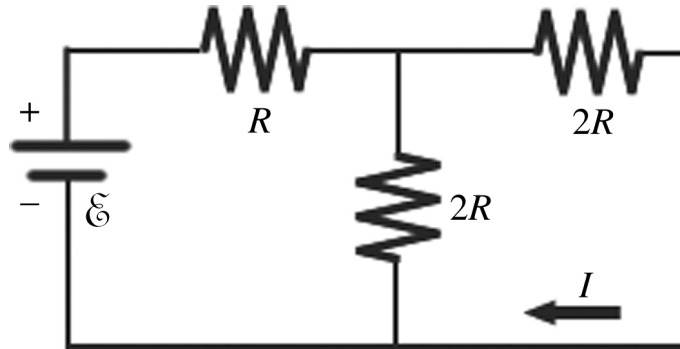
12) _____



- A) from a to b .
 B) from b to a .
 C) There is no induced current through the resistor.

13) For the circuit shown in the figure, $I = 0.50 \text{ A}$ and $R = 12 \Omega$. What is the value of the emf $\mathcal{E}$?

13) _____



- A) 24 V B) 6.0 V C) 48 V D) 12 V E) 18 V

14) A thin copper rod that is 1.0 m long and has a mass of 0.050 kg is in a magnetic field of 0.10 T. What minimum current in the rod is needed in order for the magnetic force to cancel the weight of the rod?

14) _____

- A) 4.9 A B) 1.2 A C) 7.6 A D) 9.8 A E) 2.5 A

15) If excess charge is put on a spherical solid conductor,

15) _____

- A) it remains where it was placed
 B) it spreads a little from where it was placed but not over the whole sphere
 C) it spreads uniformly over the surface of the sphere
 D) it spreads uniformly throughout the volume of the conductor

16) Which one of the following lists gives the correct order of the electromagnetic spectrum from *low to high frequencies*?

16) _____

- A) radio waves, infrared, microwaves, ultraviolet, visible, x-rays, gamma rays
 B) radio waves, ultraviolet, x-rays, microwaves, infrared, visible, gamma rays
 C) radio waves, microwaves, infrared, visible, ultraviolet, x-rays, gamma rays
 D) radio waves, microwaves, visible, x-rays, infrared, ultraviolet, gamma rays
 E) radio waves, infrared, x-rays, microwaves, ultraviolet, visible, gamma rays

17) The magnetic flux through a loop is made to vary according to the relation

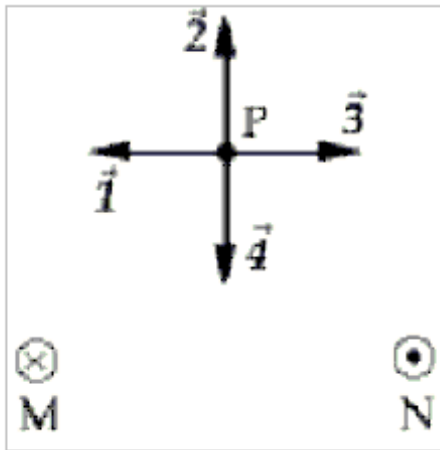
17) _____

$\phi_m = 6t^2 + 7t + 1$, where the units are SI. The EMF induced in the loop when $t = 2 \text{ s}$ is

- A) 19V B) 38V C) 31V D) 40V

- 18) Two straight wires perpendicular to the plane of this page are shown in the figure. The currents in the wires are the same. The current in M is into the page and the current in N is out of the page. The vector that represents the resultant magnetic field at point P is

18) _____

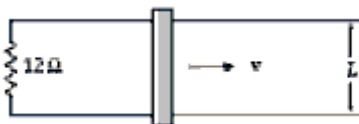


- A) 4 B) 1 C) 3 D) 2
- 19) Which one of the following types of electromagnetic wave travels through space the fastest?
- A) radio waves
B) infrared
C) microwaves
D) ultraviolet
E) They all travel through space at the same speed.

19) _____

- 20) A rod (length = 10 cm) moves on two horizontal frictionless conducting rails, as shown. The magnetic field in the region is directed perpendicularly to the plane of the rails and is uniform and constant. If a constant force of 0.60 N moves the bar at a constant velocity of 2.0 m/s, what is the current through the 12-W load resistor?

20) _____



- A) 0.32 A B) 0.37 A C) 0.43 A D) 0.39 A

ELECTRICITY AND MAGNETISM

$$|\vec{F}_E| = \frac{1}{4\pi\epsilon_0} \frac{|q_1 q_2|}{r^2}$$

$$\vec{E} = \frac{\vec{F}_E}{q}$$

$$|\vec{E}| = \frac{1}{4\pi\epsilon_0} \frac{|q|}{r^2}$$

$$\Delta U_E = q\Delta V$$

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

$$|\vec{E}| = \left| \frac{\Delta V}{\Delta r} \right|$$

$$\Delta V = \frac{Q}{C}$$

$$C = \kappa\epsilon_0 \frac{A}{d}$$

$$E = \frac{Q}{\epsilon_0 A}$$

$$U_C = \frac{1}{2} Q\Delta V = \frac{1}{2} C(\Delta V)^2$$

A = area

B = magnetic field

C = capacitance

d = distance

E = electric field

$\mathcal{E}$ = emf

F = force

I = current

ℓ = length

P = power

Q = charge

q = point charge

R = resistance

r = separation

t = time

U = potential (stored)
energy

V = electric potential

v = speed

κ = dielectric
constant

ρ = resistivity

θ = angle

Φ = flux

21)

21)

$$I = \frac{\Delta Q}{\Delta t}$$

$$R = \frac{\rho \ell}{A}$$

$$P = I \Delta V$$

$$I = \frac{\Delta V}{R}$$

$$R_s = \sum_i R_i$$

$$\frac{1}{R_p} = \sum_i \frac{1}{R_i}$$

$$C_p = \sum_i C_i$$

$$\frac{1}{C_s} = \sum_i \frac{1}{C_i}$$

$$B = \frac{\mu_0 I}{2\pi r}$$

$$\vec{F}_M = q\vec{v} \times \vec{B}$$

$$|\vec{F}_M| = |q\vec{v}| \sin \theta |\vec{B}|$$

$$\vec{F}_M = I\vec{\ell} \times \vec{B}$$

$$|\vec{F}_M| = |I\vec{\ell}| \sin \theta |\vec{B}|$$

$$\Phi_B = \vec{B} \cdot \vec{A}$$

$$\Phi_B = |\vec{B}| \cos \theta |\vec{A}|$$

$$\mathcal{E} = -\frac{\Delta \Phi_B}{\Delta t}$$

$$\mathcal{E} = B\ell v$$

Answer Key

Testname: FINAL

- 1) D
- 2) D
- 3) D
- 4) B
- 5) A
- 6) C
- 7) E
- 8) C
- 9) D
- 10) C
- 11) D
- 12) A
- 13) A
- 14) A
- 15) C
- 16) C
- 17) C
- 18) A
- 19) E
- 20) A
- 21)